Aggregates
&
Water for Concrete

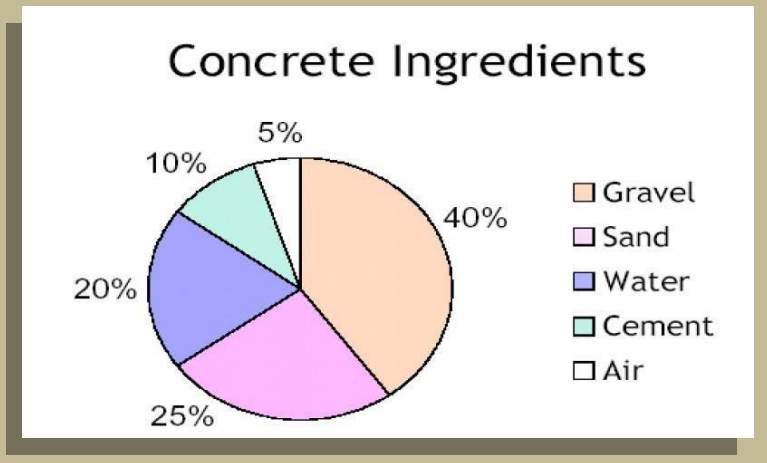




Aggregates

- **Aggregates generally occupy 65- 80% of a concrete's volume.** Aggregates are inert fillers floating in the cement paste matrix for concretes of low strength. The strength of aggregates do not contribute to the strength of concrete for low strength concrete**. The characteristics of aggregates impact performance of fresh and hardened concrete.**



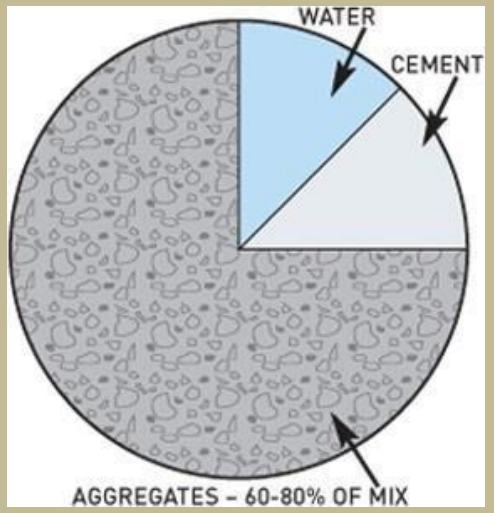


Why use aggregate

- **Reduce the cost of the concrete**
  1/4 -1/8 of the cement price
- **Reduce thermal cracking** – 100 kg of OPC

Produces about 12° C temperature rise

- **Reduces shrinkage** – 10% reduction in aggregate volume can double shrinkage
- **High aggregate** : cement ratio (A/C) desirable **A/C mainly influenced by cement content Imparts unit weight to** concrete



Aggregate Classification

- **Size:-** Coarse Aggregates & Fine Aggregates.
- **Specific Gravity:-** Light Weight, Normal Weight and Heavy Weight Aggregates.
- **Availability:-** Natural Gravel and Crushed Aggregates.
- **Shape:-** Round, Cubical, Angular, Elongated and Flaky Aggregates.
- **Texture:-** Smooth, Granular, Crystalline, honeycombed and Porous.

Aggregate Classification





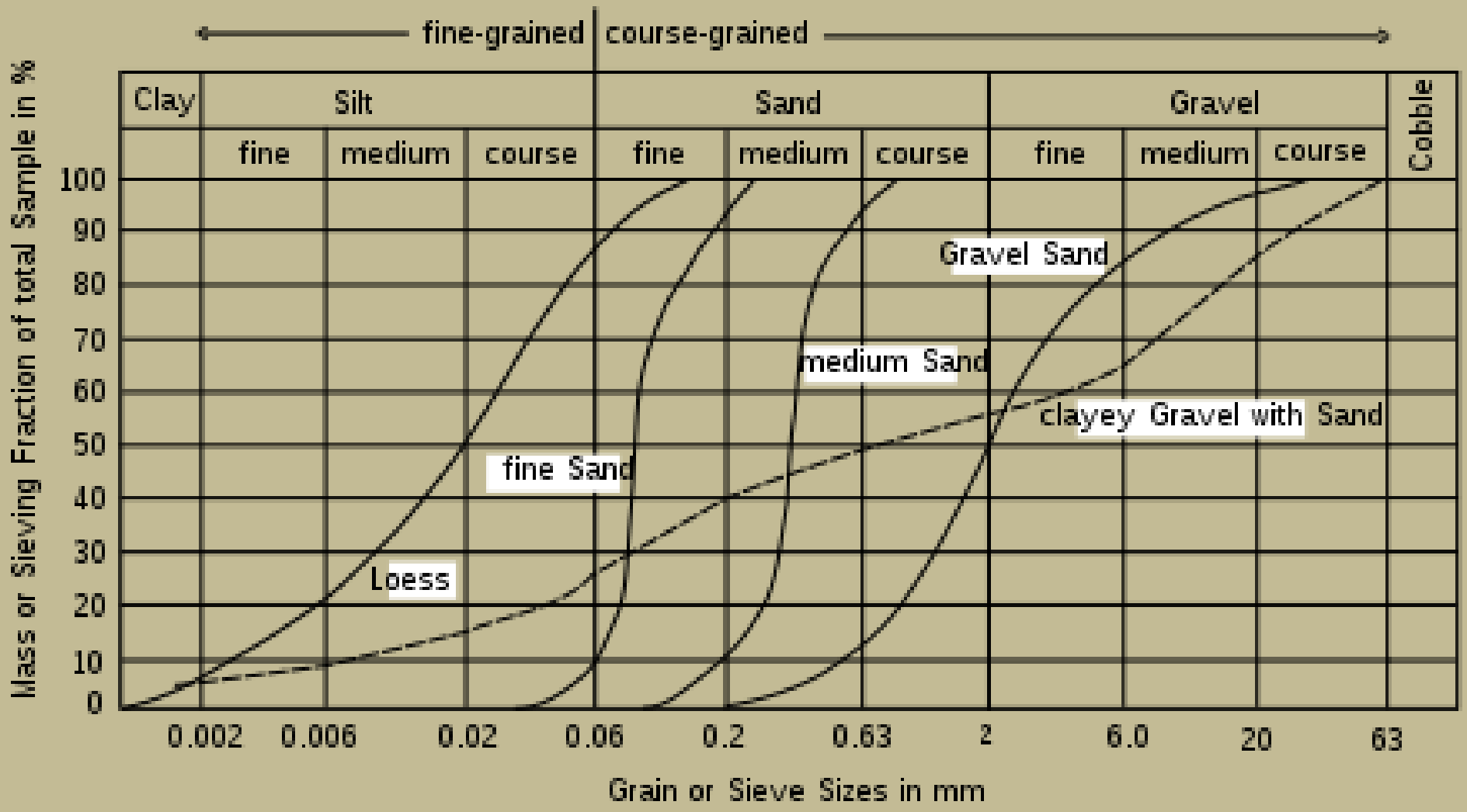
Aggregate Classification : Size



- **Fine Aggregate**
- Sand and/or crushed stone.
- < 4.75 mm.
- F.A. content usually 35% to 45% by mass or volume of total aggregate.



- **Coarse Aggregate**
- Gravel and crushed stone.
- >4.75 mm.
- Typically between 9.5 and 37.5 mm.


fine-grained
course-grained
Clay
Silt
Sand
Gravel
Cobble
fine
medium
course
fine
medium
course
fine
medium
course
100
90
80
70
60
50
40
30
20
10
0
Mass or Sieving Fraction of total Sample in %
Gravel Sand
medium Sand
clayey Gravel with Sand
fine Sand
Loess
0.002
0.006
0.02
0.06
0.2
0.63
2
6.0
20
63
Grain or Sieve Sizes in mm

Aggregate Classification : Specific Gravity

- **Normal-Weight Aggregate**
  Most common aggregates (Ex: Sand, Gravel, Crushed stone)
- Produce normal-weight concrete **2200 to 2400 kg/m^3**

- **Lightweight Aggregate**
- Expanded (Shale, Clay, Slate, Slag)
- Produce structural lightweight concrete **1350 to 1850 kg/m^3**
  And (Pumice, Scoria, Perlite, Diatomite)
  Produce lightweight insulating concrete— 250 to 1450 kg/m^3

Normal Weight Aggregates

(Ex: Sand, Gravel, Crushed stone)



Lightweight Aggregate
(Shale, Clay, Slate, Slag)





Aggregate Classification : Specific Gravity

- **Heavyweight Aggregate**
- Barite, Limonite, Magnetite, Hematite, Iron
- Produce high-density concrete up to **6400 kg/m^3**
- Used for **Radiation Shielding**







Aggregate Classification : Availability

- **Natural Gravel**

•River or seashore gravels; desert, seashore and windblown sands

- Rounded in nature
- Fully water worn or completely shaped by attrition

- **Crushed Aggregates.**
- Crushed rocks of all types; talus; screes

  Angular in nature

Aggregate Classification : Shape

- The shape of aggregates is an important characteristic since it affects the workability of concrete.



Round (spherical) concrete aggregate.

Flaky concrete aggregate.

Crushed concrete aggregate.

Aggregate Classification : Shape





Aggregate Classification : Shape

Classification	*Description*	*Examples*
Rounded	Fully water worn or completely shaped by attrition	River or seashore gravels; desert, seashore and wind-blown sands
Irregular or Partly rounded	Naturally irregular or partly shaped by attrition, having rounded edges	Pit sands and gravels; land or dug flints; cuboid rock
Angular	Possessing well-defined edges formed at the intersection of roughly planar faces	Crushed rocks of all types; talus; screes
Flaky	Material, usually angular, of which the thickness is small relative to the width and/or length	Laminated rocks

Aggregate Classification : Texture

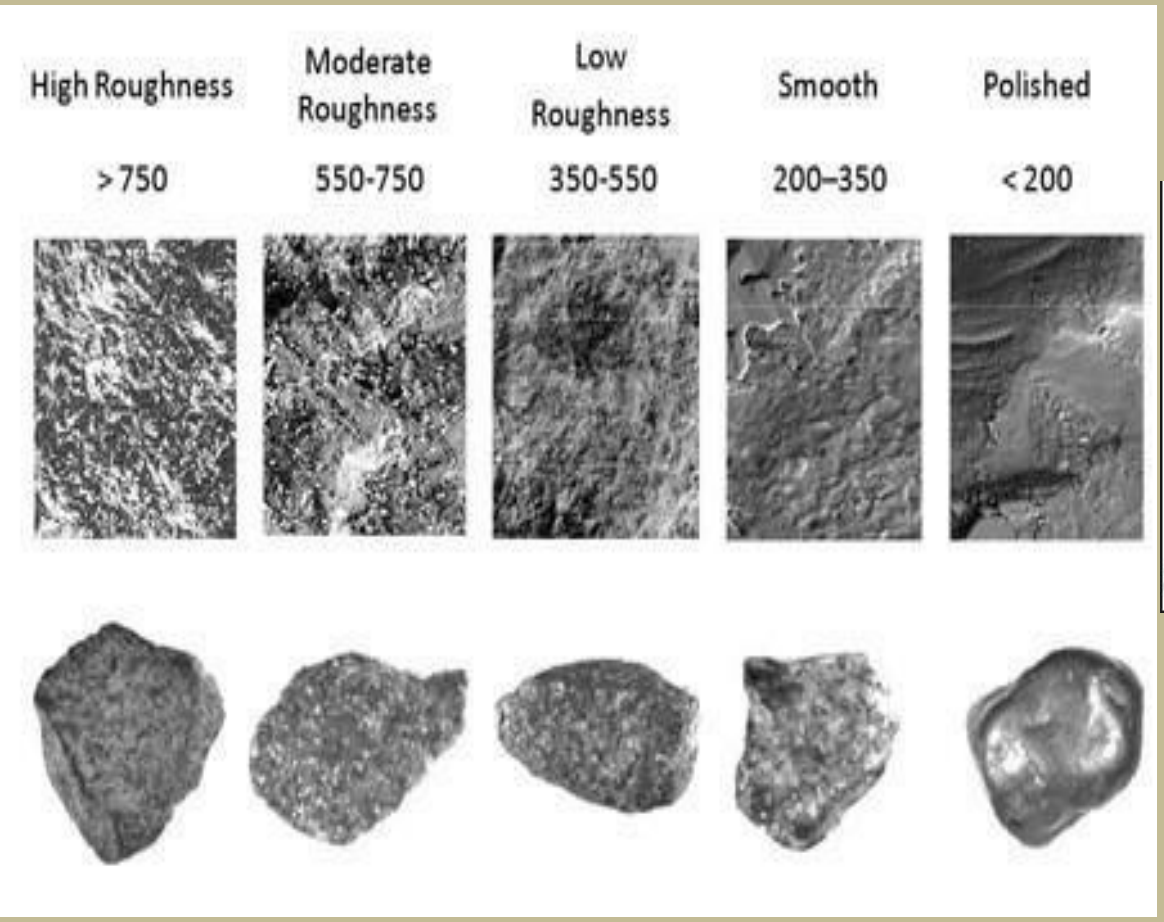
- **Surface texture is the property, the measure of which depends upon the relative degree to which particle surfaces are polished** or dull, smooth or rough.
- Surface texture depends on **hardness, grain size, pore structure, structure of the rock.**

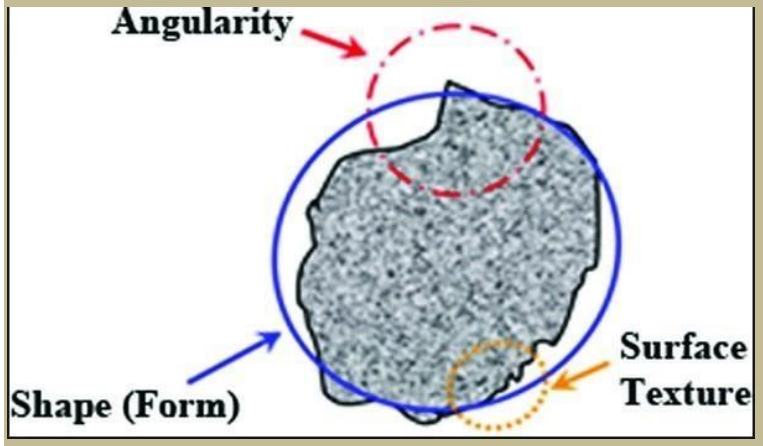
Per cent of Particles		*Water/Cement Ratio*	*Strength 28 days MPa*	
Smooth	*Rough*		*Flexural*	*Compressive*
100	0	0.54	4.3	34.8
50	50	0.57	4.6	32.1
0	100	0.60	4.8	29.5

Aggregate Classification : Texture

Group	Surface Texture	Examples
1.	Glassy	Black flint
2.	Smooth	Chert; slate; marble; some rhyolite
3.	Granular	Sandstone; oolites
4.	Crystalline	Fine : Basalt; trachyte; medium : Dolerite; granophyre; granulite; microgranite; some limestones; many dolomites. Coarse : Gabbro; gneiss; granite; granodiorite; syenite
5.	Honeycombed and porous	Scoria; Pumice, trass.

Aggregate Classification : Texture





Physical Prosperities of Aggregate : Grading

Grading is the particle-size distribution of an aggregate as determined by a sieve analysis using wire mesh sieves with square openings.

As per IS:2386(Part-1)

Fine aggregate : 6 standard sieves with openings from 150 µm to 4.75 mm. (150 µm, 300 µm, 600 µm, 1.18mm, 2.36mm, 4.75mm)

Coarse aggregate: 5 sieves with openings from 4.75mm to 80 mm. (4.75mm, 10mm, 12.5mm, 20mm, 40mm)

Physical Prosperities of Aggregate : Grading





Physical Prosperities of Aggregate : Grading

- **Grain size distribution for concrete mixes that will provide a dense strong mixture.**
- **Ensure that the voids between the larger particles are filled with medium particles.** The remaining voids are filled with still smaller particles until the smallest voids are filled with a small amount of fines.



Grading of Fine Aggregate

I.S. Sieve Designation	Percentage passing by weight for			
	Grading Zone I	Grading Zone II	Grading Zone III	Grading Zone IV
10 mm	100	100	100	100
4.75 mm	90–100	90–100	90–100	95–100
2.36 mm	60–95	75–100	85–100	95–100
1.18 mm	30–70	55–90	75–100	90–100
600 micron	15–34	35–59	60–79	80–100
300 micron	5–20	8–30	12–40	15–50
150 micron	0–10	0–10	0–10	0–15

Grading of Coarse Aggregate

IS Sieve Designation	Percentage passing for single-sized aggregate nominal size (by weight)						Percentage passing for Graded aggregate of nominal size (by weight)			
	63 mm	40 mm	20 mm	16 mm	12.5 mm	10 mm	40 mm	20 mm	16 mm	12.5 mm
80 mm	100	–	–	–	–	–	100	–	–	–
63 mm	85–100	100	–	–	–	–	–	–	–	–
40 mm	0–30	85–100	100	–	–	–	95–100	100	–	–
20 mm	0–5	0–20	85–100	100	–	–	30–70	95–100	100	100
16 mm	–	–	–	85–100	100	–	–	–	90–100	–
12.5 mm	–	–	–	–	85–100	100	–	–	–	90–100
10 mm	–	0–5	0–20	0–30	0–45	85–100	10–35	25–55	30–70	40–85
4.75 mm	–	–	0–5	0–5	0–10	0–20	0–5	0–10	0–10	0–10
2.36 mm	–	–	–	–	–	0–5	–	–	–	–

Grading of All in Aggregate

I.S. Sieve Designation	Percentage by weights passing for all in-aggragrate of	
	40 mm Nominal size	20 mm Nominal size
80 mm	100	–
40 mm	95–100	100
20 mm	45–75	95–100
4.75 mm	25–45	30–50
600 micron	8–30	10–35
150 micron	0–6	0–6

Fineness Modulus (FM)

- **The results of aggregate sieve analysis is expressed by a number called Fineness Modulus.**
- Obtained by adding the sum of the cumulative percentages by mass of a sample aggregate retained on each of a specified series of sieves and dividing the sum by 100.
- The following limits may be taken as guidance:
- Fine sand : Fineness Modulus : 2.2 - 2.6
- Medium sand : F.M. : 2.6 - 2.9
- Coarse sand : F.M. : 2.9 - 3.2

A sand having a fineness modulus more than 3.2 will be unsuitable for making satisfactory concrete.

Fineness Modulus (FM)

$$F.M. = \frac{\sum (\text{cumulative percentage retained on specified sieves})}{100}$$

where: F.M. = fineness modulus

specified sieves = 0.150 mm (No. 100), 0.30 mm (No. 50), 0.60 mm (No. 30), 1.18 mm (No. 16), 2.36 mm (No. 8), 4.75 mm (No. 4), 9.5 mm (0.375-in.), 19.0 mm (0.75-in.), 37.5 mm (1.5-in.), and larger increasing in the size ratio of 2:1.

Physical Properties of Aggregate

- **Size of aggregate**
- **Shape of aggregate**
- **Surface texture of aggregate**
- **Bulk density**
- **Specific Gravity**
- **Moisture content of aggregate**
- **Porosity and water absorption**

Soundness of Aggregate

- The ability of aggregate to resist excessive changes in volume due to changes in physical conditions.
- This occurs due to freezing and thrawing,thermal changes and alternate wetting and drying.

Physical Properties of Aggregate: Flakiness Index

- **The flakiness index of aggregate is the percentage by weight of particles in it whose least dimension (thickness)** is less than three-fifths of their mean dimension.
- **The test is not applicable to sizes smaller than 6.3 mm.**
- **The flakiness index is taken as the total weight of the material passing the various thickness gauges expressed as a percentage of the total weight of the sample taken**.
- Table 3.18 shows the standard dimensions of thickness and length gauges.

Physical Properties of Aggregate: Flakiness Index

Table 3.18. Shows Dimensions of Thickness and Length Gauges

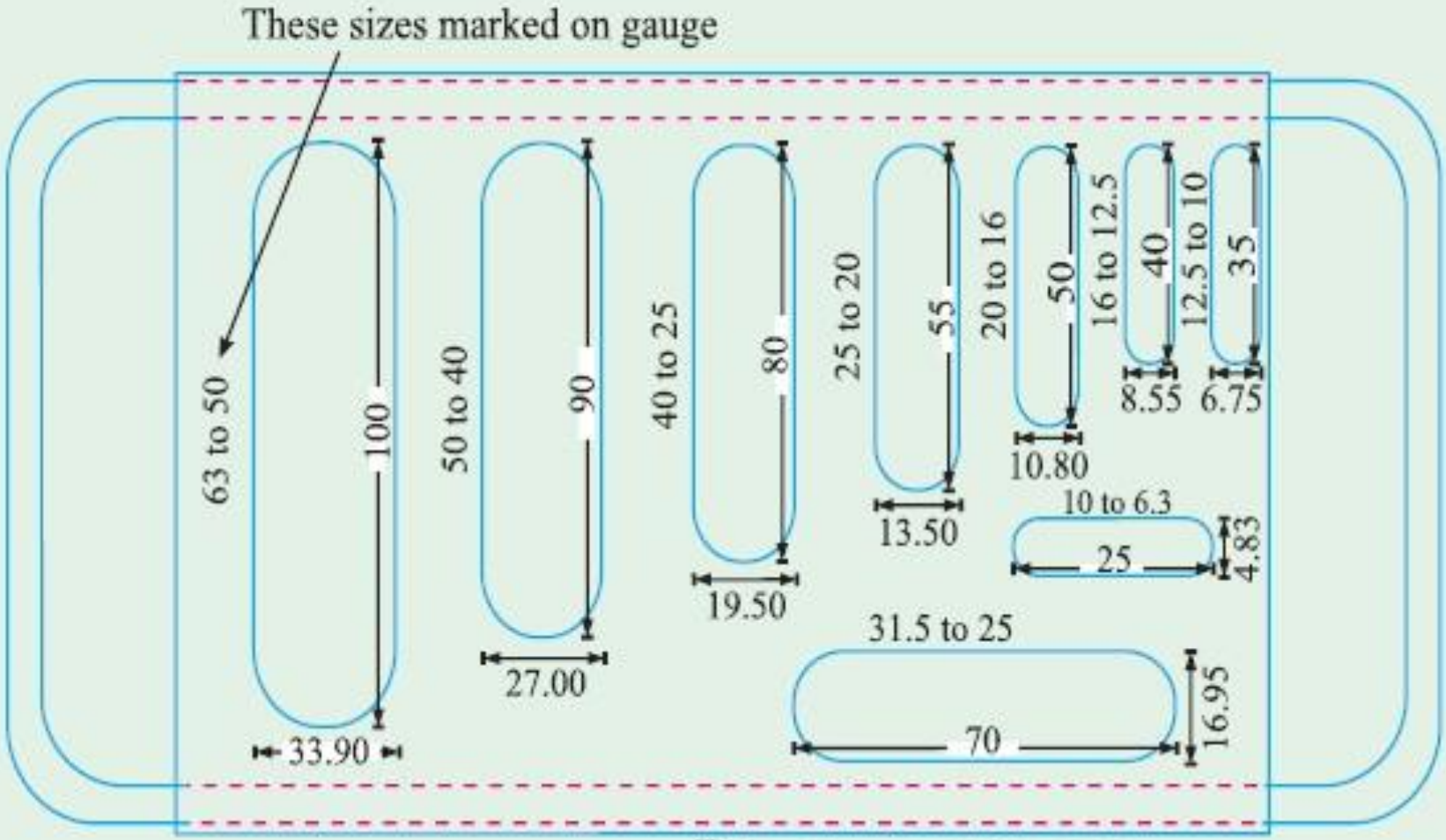
(IS: 2386 (Part I) – 1963)

Size of Aggregate Thickness		*Length of Gauge* mm*	*Gauge† mm*
Passing through IS Sieve	*Retained on IS Sieve*		
63 mm	50 mm	33.90	–
50 mm	40 mm	27.00	81.0
40 mm	25 mm	19.50	58.5
31.5 mm	25 mm	16.95	–
25 mm	20 mm	13.50	40.5
20 mm	16 mm	10.80	32.4
16 mm	12.5 mm	8.55	25.6
12.5 mm	10.0 mm	6.75	20.2
10.0 mm	6.3 mm	4.89	14.7

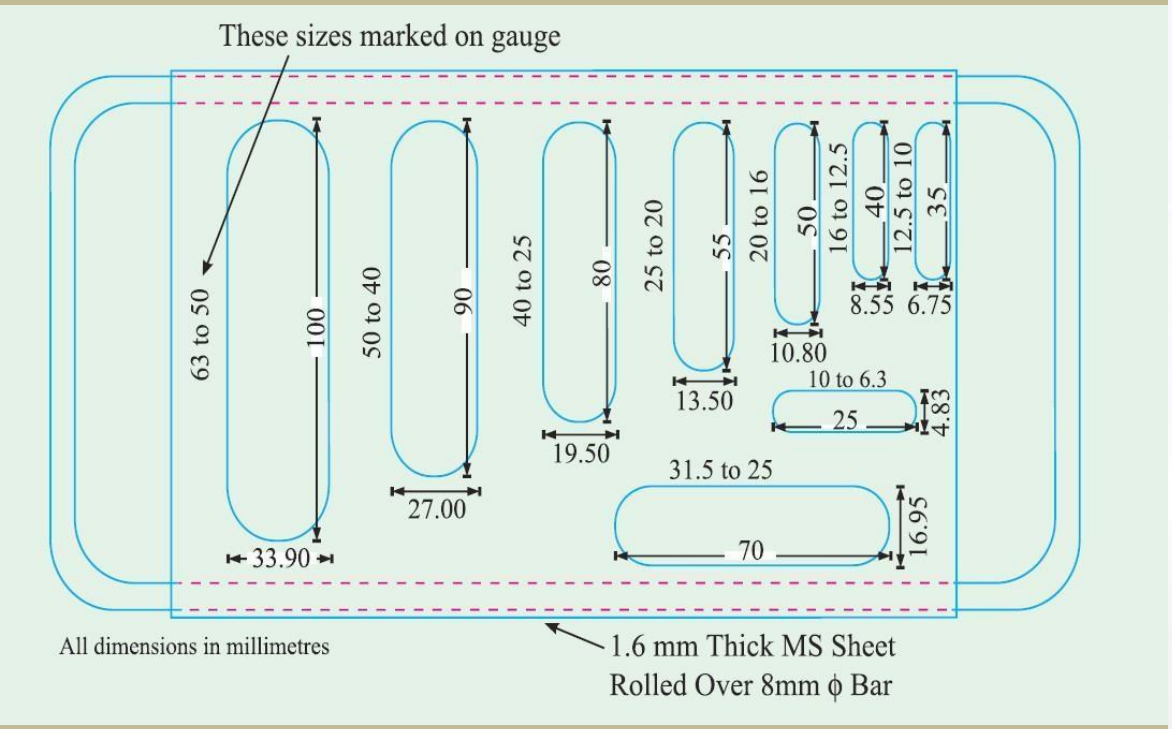
* This dimension is equal to 0.6 times the mean Sieve size.

† This dimension is equal to 1.8 times the mean Sieve size.

Physical Properties of Aggregate: Flakiness Index



Physical Properties of Aggregate: Flakiness Index

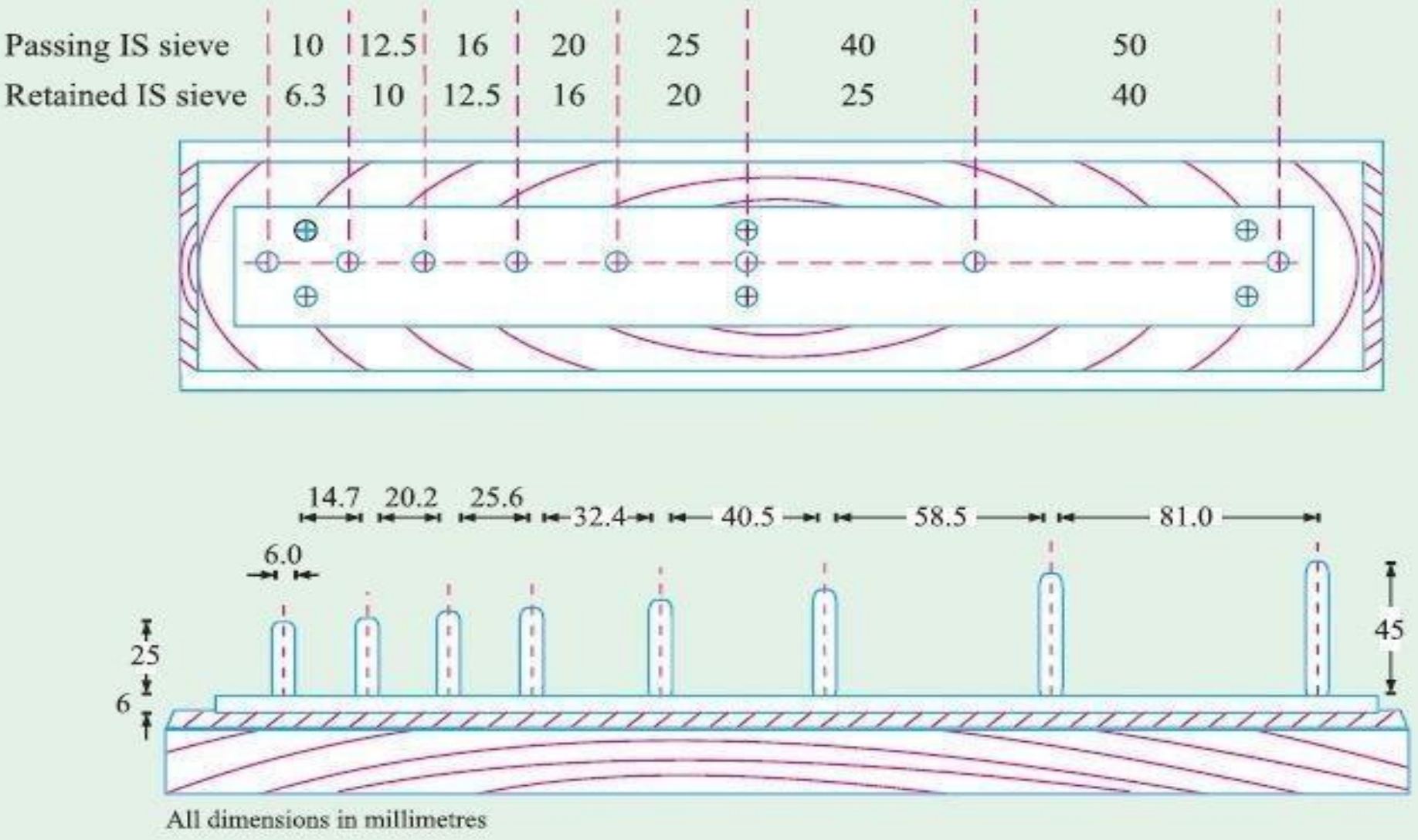




Physical Properties of Aggregate: Elongation Index

- **The elongation index on an aggregate is the percentage by weight of particles whose greatest dimension (length)** is greater than 1.8 times their mean dimension.
- The elongation index is **not applicable to sizes smaller than 6.3 mm.**
- **The elongation index is the total weight of the material retained on the various length gauges expressed as a percentage of the total weight of the sample gauged. The** presence of elongated particles in excess of 10 to 15 per cent is generally considered undesirable, but no recognized limits are laid down.

Physical Properties of Aggregate: Elongation Index



Physical Properties of Aggregate: Elongation Index





Physical Properties of Aggregate: Specific Gravity

Indian Standard Specification IS : 2386 (Part III) of 1963 gives various procedures to find out the specific gravity of different sizes of aggregates.

$$\text{Specifc Gravity} = \frac{C}{A - B}$$

$$\text{Apparent Specifc Gr ty} = \frac{C}{C - B}$$

$$\text{Water Absorption} = \frac{100(B - C)}{C}$$



A = Weight of saturated aggregate in water = (A_1 - A_2)

B = Weight of the saturated surface - dry aggregate in air

C = Weight of oven- dried aggregate in air.

A_1 = Weight of aggregate and basket in water

A_2 = Weight of empty basket in water

Physical Properties of Aggregate: Specific Gravity





Mechanical Properties of Aggregate

- Toughness
  - Resistance of aggregate to failure **by impact**.
  - Aggregate Impact Value shall not be exceed 45% by weight for aggregates used for concrete other than those used for wearing coarse and 30% for concrete for wearing surfaces.
- Hardness

Resistance to **wear and tear** and it is obtained by abrasion value.

- Crushing value of Aggregate(Compressive Strength)
  - Relative measure of the resistance of an aggregate sample to crushing under gradually applied compressive load.
  - Crushing value of aggregate is restricted to 30% for concrete used for road pavements and 45% for other structures.

Mechanical Prosperities of Aggregate : Aggregate Crushing Value

- **The "aggregate crushing value" gives a relative measure of the resistance of an aggregate** to crushing under a gradually applied compressive load.
- **The apparatus, with the test sample and plunger in position**, is placed on the compression testing machine and is loaded uniformly upto a total load of 400 kN in 10 minutes time.
- **The load is then released and the whole of the material removed from the cylinder and sieved on a 2.36 mm I.S. Sieve.**

Mechanical Prosperities of Aggregate : Aggregate Crushing Value

Mechanical Prosperities of Aggregate : Aggregate Crushing Value

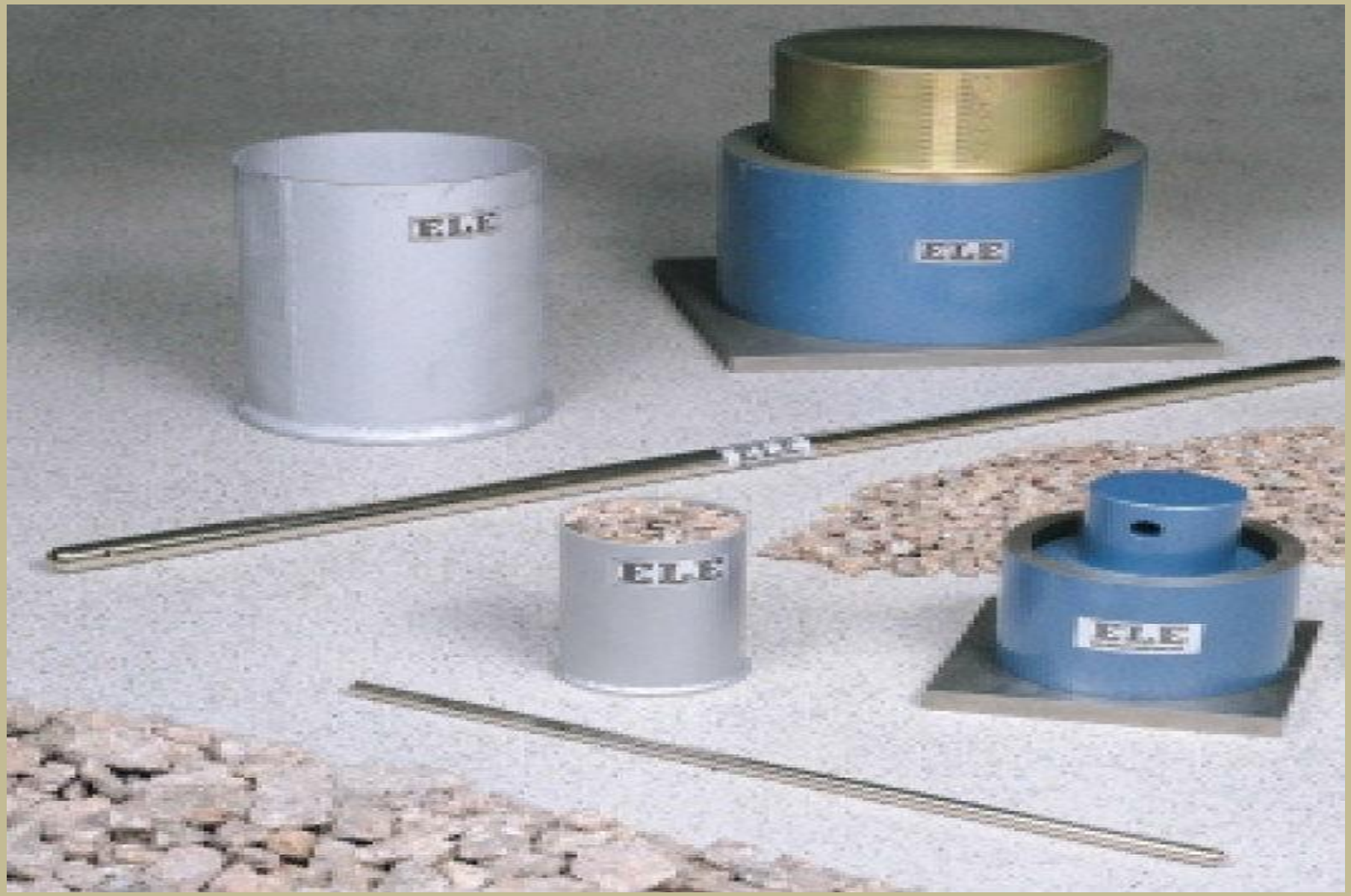
- *The aggregate crushing value* $= \frac{B}{A} \times 100$

  B = weight of fraction passing 2.36 mm sieve,

  A = weight of surface-dry sample taken in mould.

- The aggregate crushing value should not be more than 45 per cent for aggregate used for concrete other than for wearing surfaces, and 30 per cent for concrete used for wearing surfaces such a runways, roads and air field pavements.

Mechanical Prosperities of Aggregate : Aggregate Crushing Value



Mechanical Prosperities of Aggregate : Aggregate Impact Value

- **The aggregate impact value gives relative measure of the resistance of an aggregate to sudden shock or impact.**
- **The whole sample is filled into a cylindrical steel cup firmly fixed on the base of the machine. A hammer weighing about 14 kgs.** is raised to a height of 380 mm above the upper surface of the aggregate in the cup and allowed to fall freely on the aggregate.
- **The test sample shall be subjected to a total 15 such blows each being delivered at an interval of not less than one second.**

Mechanical Prosperities of Aggregate : Aggregate Impact Value



Mechanical Prosperities of Aggregate : Aggregate Impact Value

Mechanical Prosperities of Aggregate : Aggregate Impact Value

- The crushed aggregate is removed from the cup and the whole of it is sieved on 2.36 mm I.S. Sieve.
- $The\ Aggregate\ Impact\ Value = \frac{B}{A} \times 100$

  B = weight of fraction passing 2.36 mm I.S. Sieve.

  A = weight of oven-dried sample.
- The aggregate impact value should not be more than 45 per cent by weight for aggregates used for concrete other than wearing surfaces and 30 per cent by weight for concrete to be used as wearing surfaces, such as runways, roads and pavements.

Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

- Indian Standard 2386 (Part IV) of 1963 covers two methods for finding out the abrasion value of coarse aggregates: namely, by the use of Deval abrasion testing machine and by the use of Los Angeles abrasion testing machine.
- Test sample and abrasive charge are placed in the Los Angeles Abrasion testing machine and the machine is rotated at a speed of 20 to 33 rev/min.
- For gradings A, B, C and D, the machine is rotated for 500 revolutions. For gradings E, F and G, it is rotated 1000 revolutions.

Aggregate Abrasion Value Test





Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

Table 3.22. Gradings of Test Samples

Sieve Size		Weight in gm. of Test Sample For Grade						
Passing	Retained on	A	B	C	D	E	F	G
mm	mm							
80	63	-	-	-	-	2500	-	-
63	50	-	-	-	-	2500	-	-
50	40	-	-	-	-	5000	5000	-
40	25	1250	-	-	-	-	5000	5000
25	20	1250	-	-	-	-	-	5000
20	12.5	1250	2500	-	-	-	-	-
12.5	10	1250	2500	-	-	-	-	-
10	6.3	-	-	2500	-	-	-	-
6.3	4.75	-	-	2500	-	-	-	-
4.75	2.36	-	-	-	5000	-	-	-

Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

Table 3.21. Specified Abrasive Charge

Grading	*Number of spheres*	*Weight of charge (gm)*
A	12	5000 ± 25
B	11	4584 ± 25
C	8	3330 ± 20
D	6	2500 ± 15
E	12	5000 ± 25
F	12	5000 ± 25
G	12	5000 ± 25

Mechanical Prosperities of Aggregate : Aggregate Abrasion Value

- **At the completion of the above number of revolution, the material is discharged from the machine and a preliminary separation of the sample made on a sieve coarser than 1.7 mm IS Sieve.**
- **The difference between the original weight and the final weight of the test sample** is expressed as a percentage of the original weight of the test sample.
- **This value is reported as the percentage of wear**.
- **The percentage of wear should not be more than 16 per cent for concrete aggregates.**

Mechanical Prosperities of Aggregate : Aggregate Abrasion Value





Bulking of sand:

- Increase in volume of the sand due to the moisture content is known as “BULKING OF SAND”.

Reason:

- **It is due to the formation of water films around sand particles. These films of moisture exert a force known as surface tension on each particle.**
- Due to this surface tension each particle gets away from each other. Because of this no direct contact is possible among individual particles and this caused bulking of volume.
- Bulking of aggregate is dependent upon two factors,

1. Percentage of moisture content
2. Particle size of fine aggregate

- But *beyond 8-10% the film starts breaking and the volume starts reducing*.

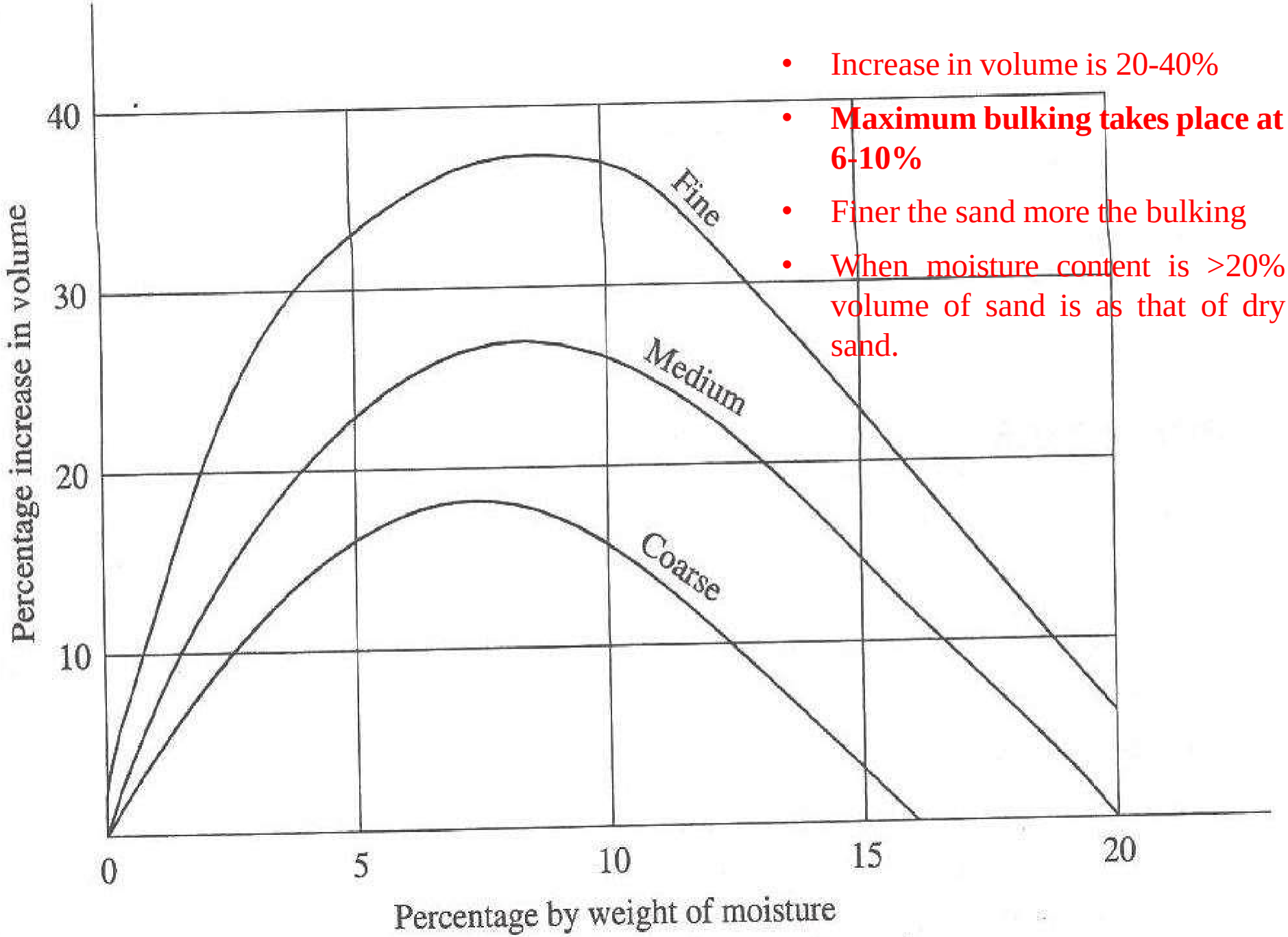


Chart showing bulking of sand

- Increase in volume is 20-40%
- **Maximum bulking takes place at 6-10%**
- Finer the sand more the bulking
- When moisture content is >20% volume of sand is as that of dry sand.

1. Cement Concrete-The "Man Made Rock"



- **Concrete is not just another material, it is 2nd most used material on the Earth**
- **It** is a composite material made up of a filler and a binder
- The binder (cement paste) "glues" the filler together to form a synthetic conglomerate.
- The constituents used for the binder are **Cement** and **Water**, while the filler can be fine and/or coarse aggregate

Binder = Cement + Water

Cement consists of Mainly

- Lime 75 to 77%
- Silica and Alumina 12 to 15%
- Iron Oxide 0.5 to 6%

Water consists of

Hydrogen and Oxygen (i.e. H_2O)

2. Requirements of water used in concrete

- Water used for mixing and curing shall be clean and free from injurious amounts of **Oils, Acids, Alkalis, Salts, Sugar,** Organic materials
- **Potable water** is generally considered **satisfactory** for mixing concrete
- Mixing and curing with **sea water** shall not be permitted.
- The **pH** value shall **not be less than 6**

2. Requirements of water used in concrete

The permissible limits for solids in water

Solids	Permissible Limits (Max)
Organic	200 mg/lit
Inorganic	3000 mg/lit
Sulphates (SO4)	500 mg/lit
Chlorides (Cl)	500 mg/lit
Suspended matter	2000 mg/lit

What if water does not meet the above requirements????

Qualities of Water

Table 4.1. Tolerable Concentrations of Some Impurities in Mixing Water

Impurity		Tolerable Concentration
Sodium and potassium carbonates and bi-carbonates	:	1,000 ppm (total). If this is exceeded, it is advisable to make tests both for setting time and 28 days strength
Chlorides	:	10,000 ppm.
Sulphuric anhydride	:	3,000 ppm
Calcium chloride	:	2 per cent by weight of cement in non-pre-stressed concrete
Sodium iodate, sodium sulphate, sodium arsenate, sodium borate	:	very low
Sodium sulphide	:	Even 100 ppm warrants testing
Sodium hydroxide	:	0.5 per cent by weight of cement, provided quick set is not induced.

Qualities of Water

Salt and suspended particles	:	2,000 ppm. Mixing water with a high content of suspended solids should be allowed to stand in a s ettling basin before use.
Total dissolved salts	:	15,000 ppm.
Organic material	:	3,000 ppm. Water containing humic acid or such organic acids may adversely affect the hardening of concrete; 780 ppm. of humic acid are reported to have seriously impaired the strength of concrete. In the case of such waters there- fore, further testing is necessary.
pH	:	shall not be less than 6

Qualities of Water

Table 4.2. Permissible limit for solids as per IS 456 of 2000

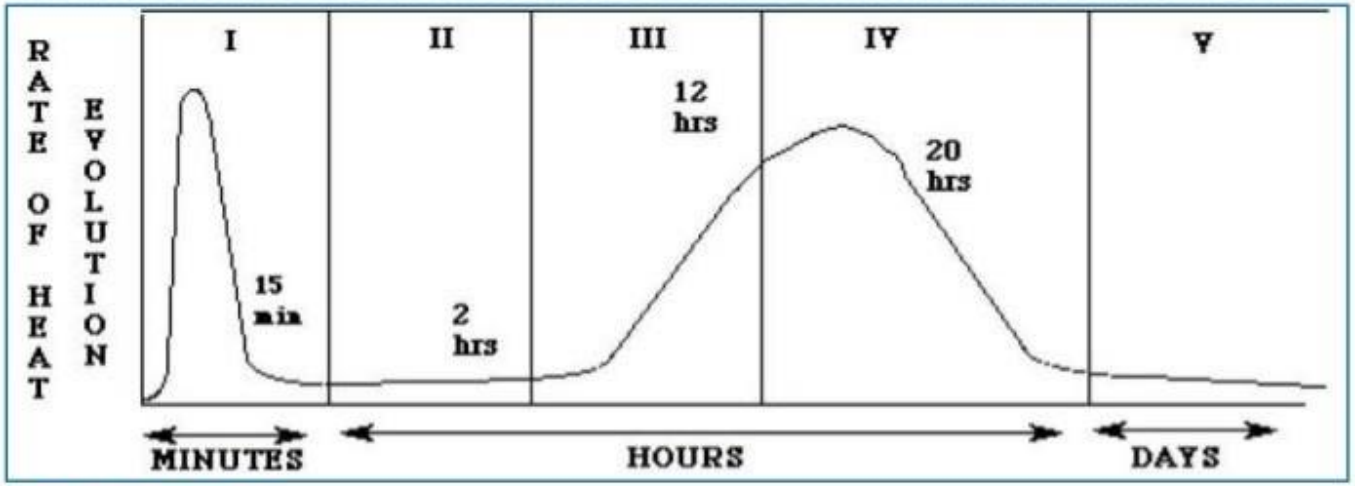
Material	Tested as per	Permissible limit Max.
Organic	IS 3025 (pt 18)	200 mg/l
Inorganic	IS 3025 (pt 18)	3000 mg/l
Sulphates (as So_3)	IS 3025 (pt 24)	400 mg/l
Chlorides (as Cl)	IS 3025 (pt 32)	2000 mg/l for concrete work not containing embedded steel and 500 mg/l for reinforced concrete work
Suspended	IS 3025 (pt 17)	2000 mg/l

3. Hydration

- Concrete achieves its **strength** through a chemical process called **Hydration**.

- **Hydration** is a complex process but in simple terms, is the **reaction between water and the cement** in the mix.

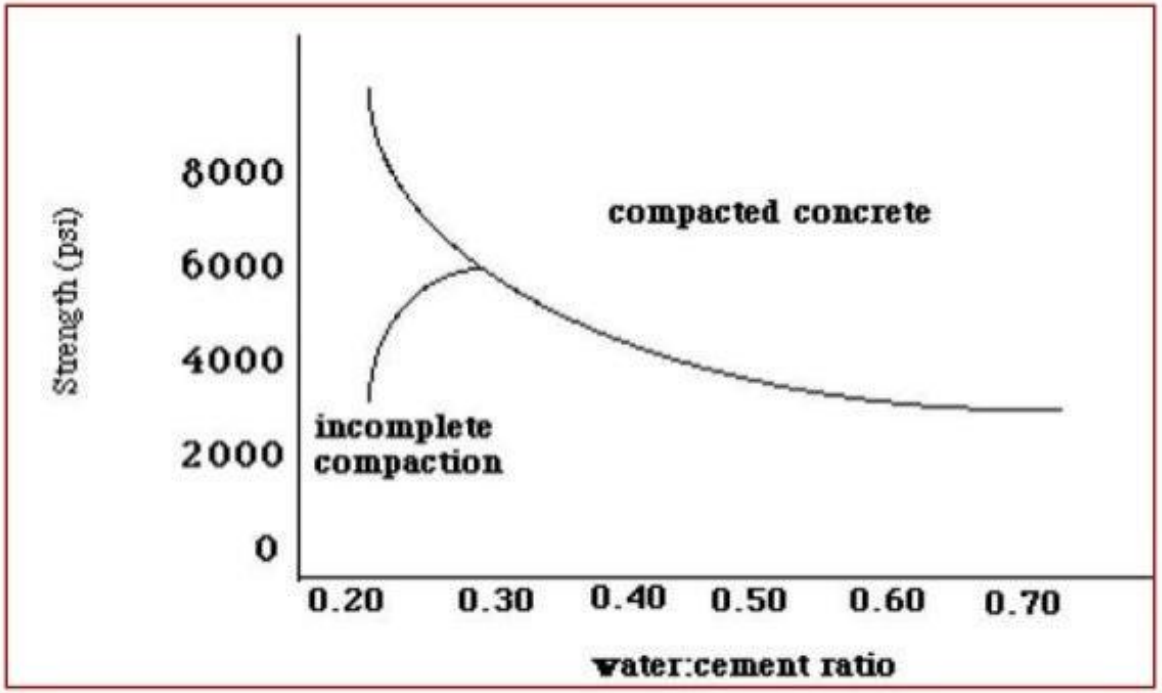
3. Hydration



- **Stage I HYDROLYSIS** of the cement compounds occurs rapidly with a temperature increase of several degrees
- **Stage II** is known as the **DORMANCY PERIOD.** The evolution of heat slows dramatically in this stage. This can last from one to three hours. here concrete is in a plastic state which allows it to transport and place without major difficulty. At the end of this stage initial setting begins
- **In stages III and IV**, the concrete starts to **HARDEN** and the heat evolution increases due primarily to the hydration of tricalcium silicate
- **Stage V** is reached after 36 hours. The slow formation of hydrate products occurs and continues as long as water and unhydrated silicates are present.

4. Water/Cement Ratio and Strength

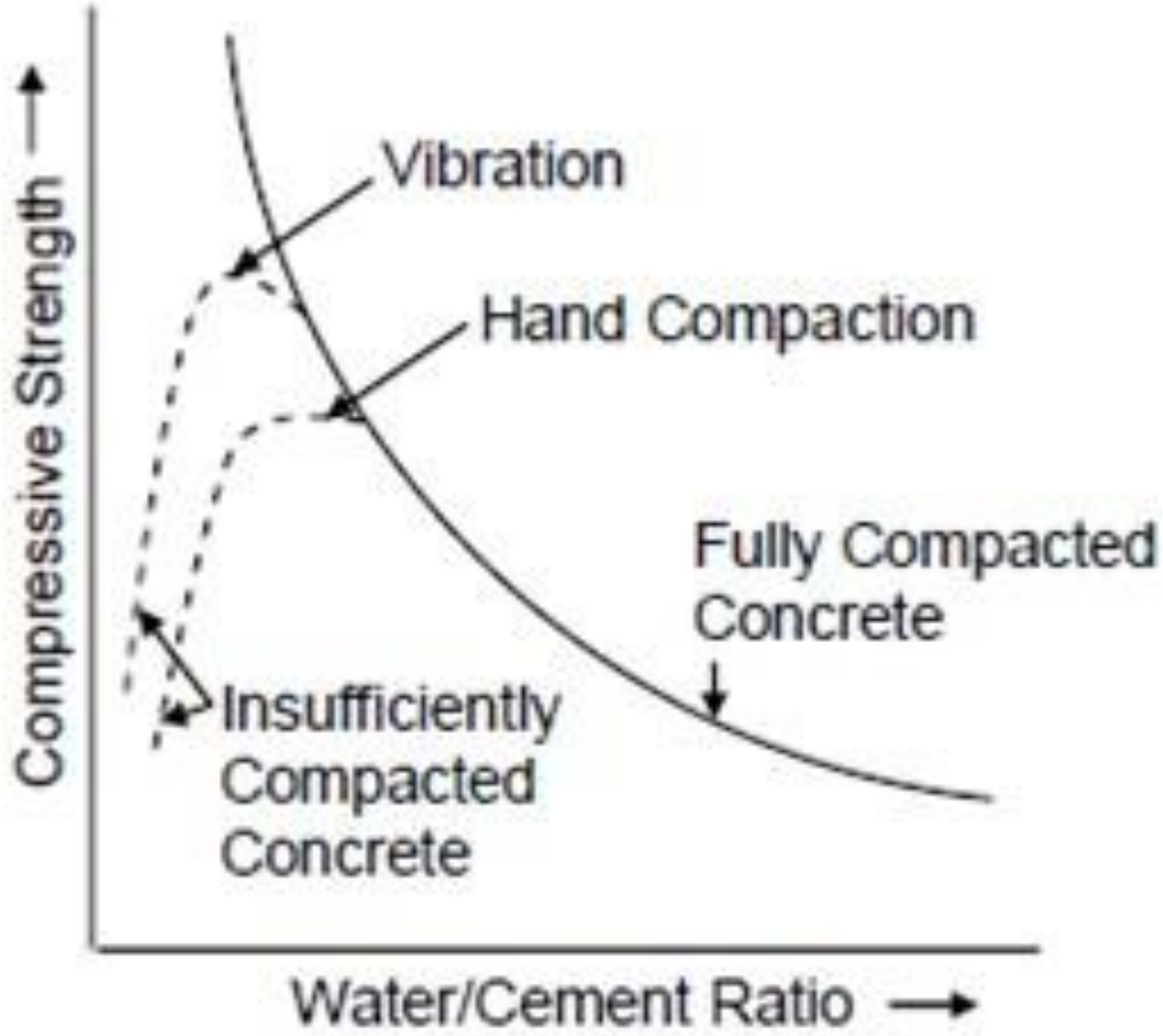
- The most important indicator of **strength**
- Lower w/c ratio is, the higher the final concrete strength
- Concept was developed by ***Duff Abrams*** *of The PCA* in the early 1920's



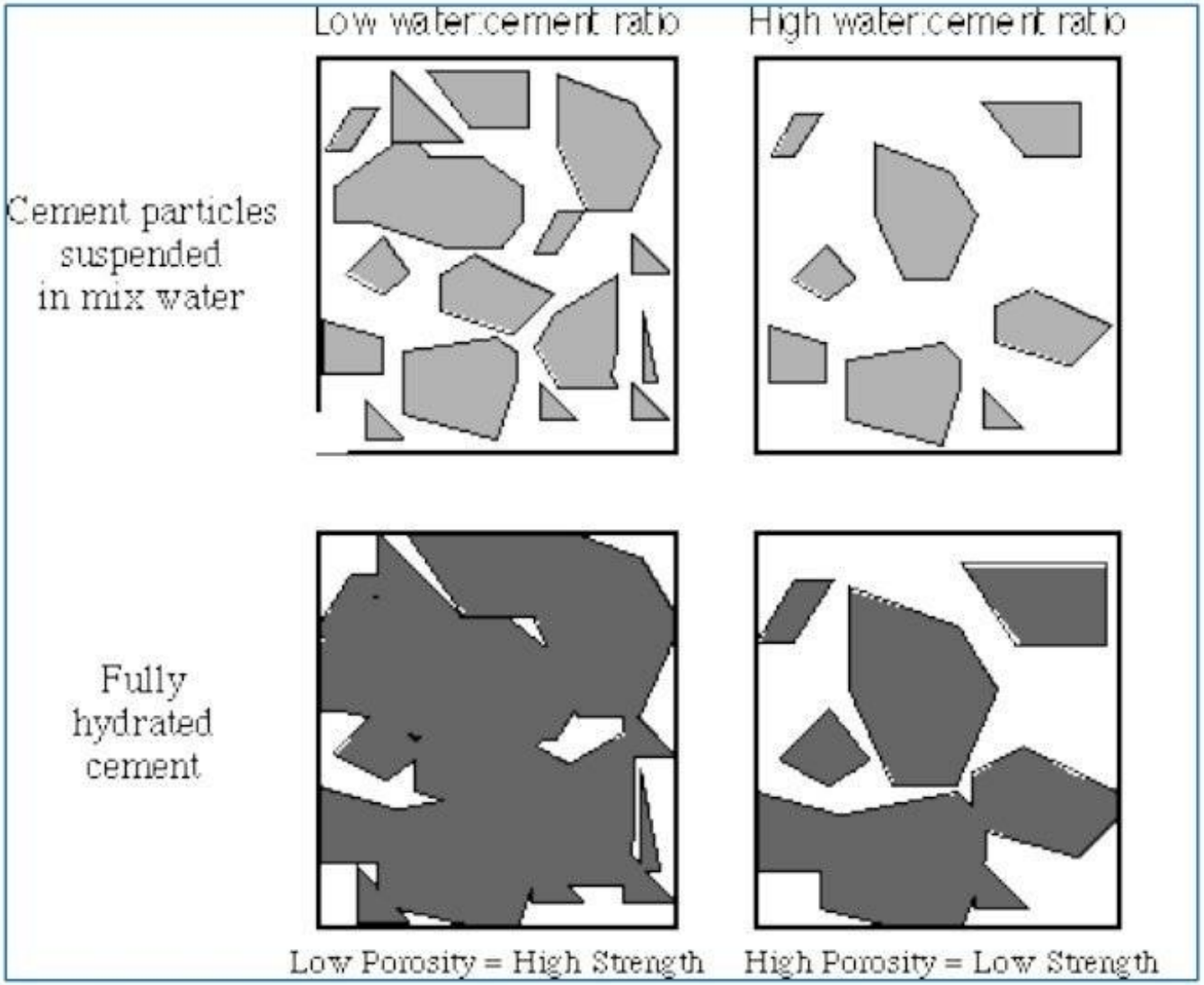
Water-Cement Ratio

The most important property of hydrating cement Water needed for hydration, w/c = 0.22-0.25 Extra water is needed for workability but causes voids

- Decreases strength
- Decreases durability
- Decreases bond between successive layers
- Decreases bond between concrete and rebar
- Increases permeability
- Increases volume change from wetting and drying


Compressive Strength
Vibration
Hand Compaction
Fully Compacted Concrete
Insufficiently Compacted Concrete
Water/Cement Ratio

4. Water/Cement Ratio and Strength



4. Water/Cement Ratio and Strength

(w/c) Ratio	0.40	0.50	0.60	0.70	0.80
Probable Strength(%)	100	87	70	55	44

Factors	**Low w/c ratio**	**High w/c ratio**
Strength	High	Low
Permeability	Low	High
Shrinkage	Low	High

Adding extra water to concrete!!!

- Adding more water creates a diluted paste that is weaker and more susceptible to cracking and shrinkage
- Shrinkage leads to micro-cracks (zones of weakness)
- Once the fresh concrete is placed, excess water is squeezed out of paste by weight of aggregate and cement
- The excess water bleeds out onto the surface.
- The micro channels and passages that were created inside the concrete to allow that water to flow become weak zones

Adding extra water to concrete!!!

- This affects the compressive, tensile and flexural strengths, the porosity and the shrinkage
- **Loss** of Inherent good qualities like **Cohesiveness** and **Homogeneity**
- **Harmful** to **Strength** and **Durability**
- Sowing the seed of **Cancer** in concrete
- It is an **Abuse**
- It is a **Criminal act**
- **Un-engineering** ----------(M.S.Shetty, Eminent Author)

Advantages of low water/cement ratio

* Increased strength.
* Lower permeability.
* Increased resistance to weathering.
* Better bond between concrete and reinforcement.
* Reduced drying shrinkage and cracking.
* Less volume change from wetting and drying.

5. Workability

- The ease with which freshly mixed concrete can be transported, placed and finished without segregation

- Influencing factors
  - Size, Shape, Texture and grading of aggregate
  - **Water Content**
  - **Admixtures**
    - **Mineral (Fly ash, Silica fume, GGBFS)**
    - **Chemical**
    - **Air entraining**



Admixture types

- Chemical and Air-entraining admixtures are Covered by IS:9301-1999
  - a) Accelerating admixtures
  - b) Retarding Admixtures
  - c) **Water-reducing admixtures (plasticizers)**
  - d) **Air-entraining admixtures and**
  - e) **Super-plasticizing admixtures**

Water-reducing admixtures

- An admixture which either increases workability of freshly mixed mortar or concrete without increasing water content or maintains workability with a reduced amount of water
- Role of water reducers is to deflocculate the cement particles agglomerated together and release the water tied up in these agglomerations
- Can be categorized according to their active ingredients
  - salts and modifications of hydroxylized carboxylic acids (HC type)
  - salts and modifications of lignosulfonic acids and
  - Polymeric materials (PS type)
- Reduces water demand 7-10%

Air-entraining admixtures

- Which causes air to be incorporated in the form of minute bubbles in the concrete or mortar during mixing, usually to increase workability and resistance to freezing and thawing and disruptive action of de-icing salts
- Reduces bleeding and segregation of fresh concrete
- Can be categorized into four groups:
  - salts of wood resins
  - **synthetic detergents**
  - salts of petroleum acids,
  - fatty and resinous acids and their salts

Super-plasticizing admixtures

- Which imparts very high workability or allows a large decrease in water content for a given workability
- Reduce water content by 12 to 30 percent
- The effect of superplasticizers lasts only 30 to 60 minutes and is followed by a rapid loss in workability
- Superplasticizers are usually added to concrete at the jobsite
- Example : Glenium-BASF, Supaflo-Don Chemicals

6. Durability of Concrete

The factors influencing durability include

- The environment
- The cover to embedded steel
- The type and quality of constituent materials
- **The cement content and water/cement ratio**
- Workmanship to obtain full compaction
- **Efficient curing**
- The shape and size of the member

6. Durability of Concrete

- **Cement content and water/cement ratio For bridges with individual span lengths more than 30 m**

Structural Member	Min. cement content for all Exposure conditions (kg/cu.m,)	Max. water cement ratio	
		Exposure conditions	
		Normal	Severe
PCC members	360	0.45	0.45
RCC members	400	0.45	0.40
PSC members	400	0.40	0.40

6. Durability of Concrete

Minimum Cement content and Maximum water/cement ratio for Culverts and other incidental construction

Structural Member	Min. cement content (kg/cu.m,)		Max. water cement ratio	
	Exposure conditions		Exposure conditions	
	Normal	Severe	Normal	Severe
PCC members	250	310	0.50	0.45
RCC members	310	400	0.45	0.40

6. Durability of Concrete

- The minimum cement content is based on 20 mm aggregate For larger size aggregates, it may be reduced up to 10%
- For underwater concreting, the cement content shall be increased by 10%
- Severe conditions of exposure shall mean alternate wetting and drying due to sea spray, alternate wetting and drying combined with freezing and buried in soil having corrosive effect.

6. Durability of Concrete- Curing

Curing methods

1. Water curing 2. Steam curing 3. Curing compounds

Water curing

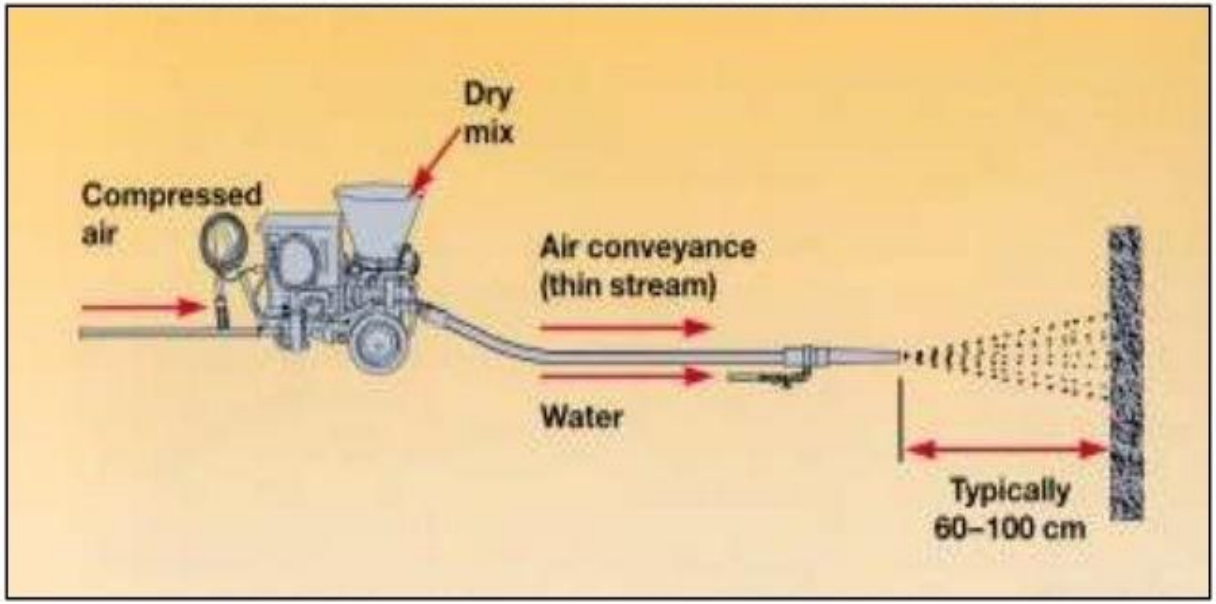
- Sea water shall not be used for curing
- Seawater shall not come into contact with concrete members unless it has attained adequate strength
- Exposed surface of concrete shall be kept continuously in a damp or wet condition by **ponding or by covering with a layer of sacks, canvas, Hessian** or similar materials and shall be kept constantly wet for a period of not less than 14 days from the date of placing of concrete.

6. Durability of Concrete- Curing

Curing compounds

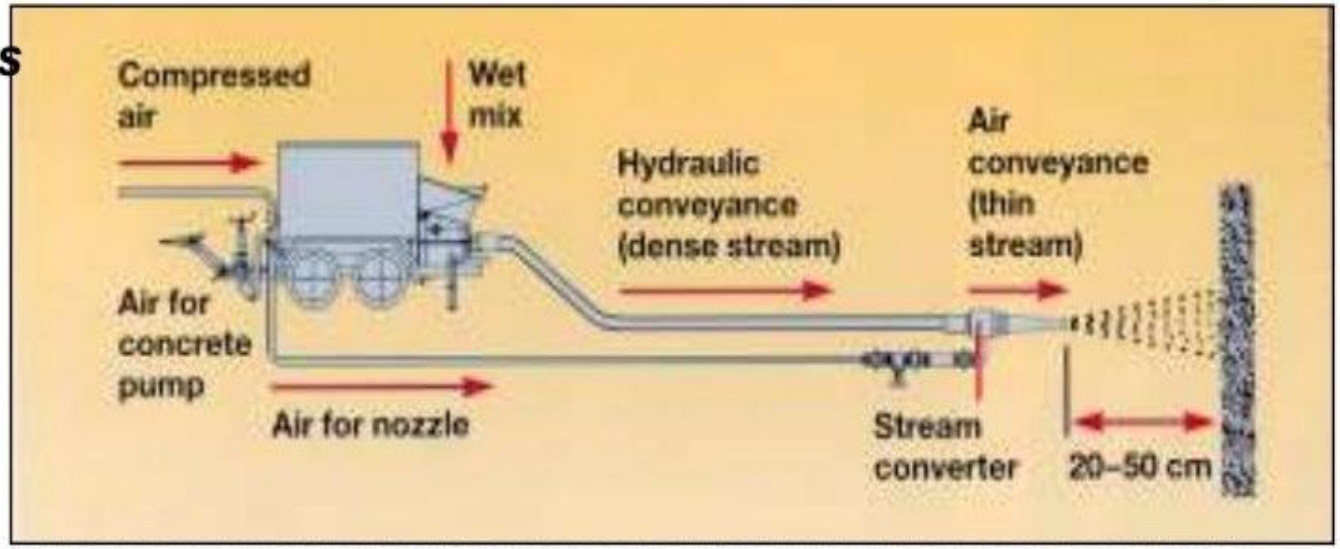
- All concrete cured by this method shall receive two applications of the curing compound
- The first coat shall be applied immediately after acceptance of concrete finish
- The second application shall be made after the first application has set
- Curing compounds shall not be used on any surface which requires further finishing to be applied
- No curing compound shall be permitted in locations where concrete surfaces are required to be bonded together

7. Sprayed Concrete



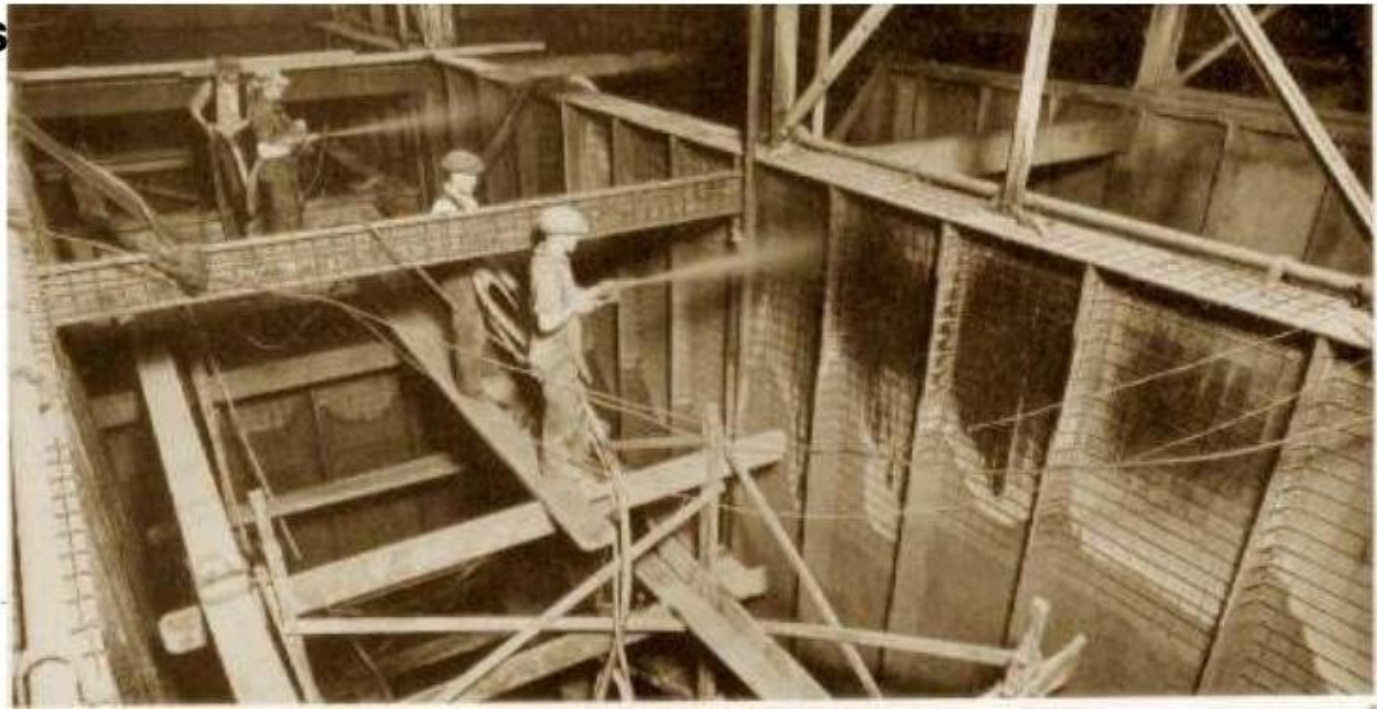
"Dry spray" process

"wet spray" process



Properties and advantages of sprayed concrete over poured concrete

- Low Water/Cement Ratio
- High Strengths with Rapid Strength Gain
- High Density/Low Permeability
- Enhanced Adhesion and Bond Strength
- High Speed High Output
- Reduction in Formwork Costs
- Ease of Access



TYPICAL APPLICATIONS AND USES

New Construction

Shell roofs and domes
Retaining walls
Piled wall facings
Silo structures
Barrel vaulting
Diaphragm walls
Caissons
Blast proof structures
Bank vaults

Underground Construction

Tunnel linings
Storage reservoirs

TYPICAL APPLICATIONS AND USES

Water Retaining Structures

- Sea and river walls
- Reservoirs and dams
- Aqueducts
- Swimming pools
- Water towers
- Canal linings
- Irrigation and drainage channels

Protective Coatings

- Fire protection to structural steelwork
- Refractory linings
- Pipeline encasement
- Rock and soil stabilisation

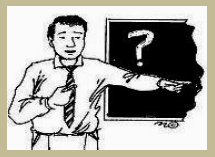
TYPICAL APPLICATIONS AND USES

Strengthening and Repair

- Concrete damaged by reinforcement corrosion
- Fire damaged structures
- Cooling towers
- Bridges
- Jetties and wharves
- Brick arches and tunnels
- Tunnel linings
- overlays

Free Formed Structures

- Swimming pools
- Landscaping
- Climbing walls
- Theme parks
- Sculpture
- Water sports slalom courses
- Bobsleigh runs
- Zoological structures



Any ?



When it rains – look for the rainbows – when it's *dark*, *look for the stars*.

